

Assignment - 4

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Section : 23

Course Code : CSE422

Answer the Question no-1

a) Marginal Probability of Teacher,

$$P(\text{Teacher}) = 0.15 + 0.05 + 0.18$$

$$= 0.53$$

(Answer)

$$b) P(\text{Low} | \text{Mobile}) = \frac{P(\text{Mobile} | \text{Low})}{P(\text{Mobile})}$$

$$= \frac{0.07 + 0.15}{0.2 + 0.07 + 0.18 + 0.15}$$

$$= 0.3667$$

(Answer)

$$c) P(\text{Student} | \text{Mobile}) = \frac{0.2 + 0.07}{0.6} = 0.45$$

$$P(\text{Low} | \text{Mobile}) = 0.3667$$

$$\therefore P(\text{Student} | \text{Mobile}) \times P(\text{Low} | \text{Mobile}) \\ = 0.45 \times 0.3667 = 0.165$$

$$P(\text{Student Low} | \text{Mobile}) = \frac{0.07}{0.6} = 0.1167$$

$$\therefore 0.1167 \neq 0.165$$

$\therefore$ Not conditionally independent

Answer the Question no-2

$$\text{Given, } P(S) = 0.15 = \frac{1}{6}$$

$$P(L) = 0.85$$

$$P(MIS) = 0.9$$

$$P(MIL) = 0.04$$

$$P(L|M) = \frac{P(MIL) \cdot P(L)}{P(MIS) \cdot P(S) + P(MIL) \cdot P(L)} \\ = \frac{0.04 \times 0.85}{(0.9 \times 0.15) + (0.04 \times 0.85)}$$

$$= 0.20$$

$$\therefore P(\text{Not Spam} | \text{Marked as Spam}) = 0.20$$

Answer to the Question No-3

$$P(\text{Yes}) = \frac{4}{8} = 0.5$$

$$P(\text{No}) = \frac{4}{8} = 0.5$$

$$P(\text{Evening} | \text{Yes}) = \frac{1}{4} = 0.25$$

$$P(\text{Lakeside} | \text{Yes}) = \frac{2}{4} = 0.5$$

$$P(\text{Sound} = \text{Yes} | \text{Yes}) = \frac{4}{4} = 1$$

$$P(\text{Evening} | \text{No}) = \frac{1}{4} = 0.25$$

$$P(\text{Lakeside} | \text{No}) = \frac{0}{4} = 0$$

$$P(\text{Sound} = \text{Yes} | \text{No}) = \frac{1}{4} = 0.25$$

For Rate = Yes,

$$P(\text{Yes} | x) \propto 0.5 \times 0.25 \times 0.5 \times 1 = 0.0625$$

For Rare = No ,

$$P(\text{No} | x) \propto 0.5 \times 0.25 \times 0 \times 0.25 = 0$$

Final prediction $0.0625 > 0$,

$\therefore$ Rare Species = Yes

Answer the Question no-4

a) Using multinomial Naïve Bayes,

$$P(y | x_1, x_2) \propto P(y) \times P(x_1 | y) \times P(x_2 | y)$$

$$P(y=0) = \frac{3}{6}$$

$$P(y=1) = \frac{2}{6}$$

$$P(y=2) = \frac{1}{6}$$

For $y=0$,

$$P(x_1 | y=0) = \frac{1}{3}$$

$$P(x_2 | y=0) = \frac{2}{3}$$

$$\begin{aligned} \text{score}(y=0) &= \frac{3}{6} \times \frac{1}{3} \times \frac{2}{3} \\ &= \frac{1}{9} = 0.111 \end{aligned}$$

For $y=1$,

$$P(x_1 | y=1) = \frac{1}{2}$$

$$P(x_2 | y=1) = \frac{1}{2}$$

$$\text{score}(y=1) = \frac{2}{6} \times \frac{1}{2} \times \frac{1}{2} = 0.083$$

For $y=2$,

$$P(x_1 | y=2) = \frac{1}{2}$$

$$P(x_2 | y=2) = \frac{1}{2}$$

$$\text{score}(y=2) = \frac{1}{6} \times \frac{1}{2} \times \frac{1}{2} = 0.0417$$

$$\therefore \text{score}(y=0) > \text{score}(y=1) > \text{score}(y=2)$$

$$\therefore \text{class, } y=0$$

b) If a new continuous feature $x_3 \in \mathbb{R}$ is added, we can use Gaussian Naïve Bayes to that feature,

$$P(x_3|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} e^{-\frac{(x_3 - \mu_y)^2}{2\sigma_y^2}}$$

∴ Final classifier,

$$P(y|x_1, x_2, x_3) \propto P(y) P(x_1|y) P(x_2|y) P(x_3|y)$$

Therefore, use multinomial Naïve Bayes for x_1, x_2 and Gaussian Naïve Bayes for continuous.

Answer the Question no-5

a)

x	y	$\hat{y} = 2x + 1$	Error $\hat{y} - y$	Squared Error
1	5	3	-2	4
3	12	7	-5	25
5	15	11	-4	16

$$\therefore \text{MSE} = \frac{4 + 25 + 16}{3} = 15$$

b) Learning rate, $\alpha = 0.05$

Iteration 1,

$$\frac{\partial \text{MSE}}{\partial m} = \frac{2}{3} [(-2)(4) + (-5)(3) + (-4)(5)]$$
$$= -24.667$$

$$\frac{\partial \text{MSE}}{\partial b} = \frac{2}{3} [-2 - 5 - 4] = -7.33$$

update, $m = 2 - 0.05(-24.67) = 3.23$

$$b = 1 - 0.05(-7.33) = 1.367$$

Iteration 2,

$$\frac{\partial \text{MSE}}{\partial m} = \frac{2}{3} [(-0.4)(1) + (-0.933)(3) + (2.533)(5)]$$
$$= 6.311$$

$$\frac{\partial \text{MSE}}{\partial b} = \frac{2}{3} [-0.4 - 0.93 + 2.53]$$
$$= 0.800$$

update, $m = 3.233 - 0.05 \times 6.31$

$$= 2.918$$

$$b = 1.367 - 0.05 \times 0.8 = 1.327$$

Answer the Question no-6,

a) initial values,

$$w_1 = 0.5$$

$$w_2 = 0.4$$

$$b = 0.3$$

$$\alpha = 0.05$$

$$\hat{y} = w_1 x_1 + w_2 x_2 + b$$

Initial MSE,

<u>Sample</u>	<u>Actual y</u>	<u>Predicted $\hat{y}$</u>	<u>Error</u>
1	5	2.0	-3.0
2	4	2.1	-1.9
3	1	1.9	0.9
4	9	2.2	-6.8

$$MSE = \frac{(-3)^2 + (-1.9)^2 + (0.9)^2 + (-6.8)^2}{4}$$

$$= 14.915$$

b) Iteration 1,

$$\frac{\partial \text{MSE}}{\partial w_1} = -13.6$$

$$\frac{\partial \text{MSE}}{\partial w_2} = -8.0$$

$$\frac{\partial \text{MSE}}{\partial b} = -5.4$$

Updated values,

$$w_1 = 0.5 - 0.05(-13.6) = 1.18$$

$$w_2 = 0.4 - 0.05(-8.0) = 0.80$$

$$b = 0.3 - 0.05(-5.4) = 0.57$$

Iteration 2,

$$\frac{\partial \text{MSE}}{\partial w_1} = -6.03$$

$$\frac{\partial \text{MSE}}{\partial w_2} = 2.75$$

$$\frac{\partial \text{MSE}}{\partial b} = -0.82$$

Updated values,

$$w_1 = 1.18 - 0.05 \times (-6.03) = 1.48$$

$$w_2 = 0.8 - 0.05 \times 2.75 = 0.6625$$

$$b = 0.57 - 0.05(-0.82) = 0.611$$

~~$w_1 = 1.4815$~~

w_2

c) $x_1 = 2.5, x_2 = 2$

$$\hat{y} = 1.48 \times 2.5 + 0.66 \times 2 + 0.611$$

$$= 5.639$$

$$\approx 5.64$$

∴ predicted test score = 5.64

$$[(\hat{y}-1) \log(\hat{y}-1) + \hat{y} \log \hat{y}] \sum \frac{1}{n} = 7.29$$

$$PFC.L =$$

Answer the Question no-7

a) Given, $X = [0.2, 0.4, 0.8]$, $Y = [0, 0, 1]$

$Y \in \{0, 1\}$

∴ Its a Binary classification problem.

∴ Suitable loss function : Binary cross Entropy (BCE)

b)

x	y	$z = wx + b$	$\hat{y} = \sigma(z)$
0.2	0	1.6	0.83
0.4	0	2.2	0.9
0.8	1	3.4	0.96

$$BCE = -\frac{1}{n} \sum [y \log \hat{y} + (1-y) \log (1-\hat{y})]$$

$$= 1.374$$

$$c) \frac{\partial L}{\partial w} = \frac{1}{n} \sum (\hat{y} - y) x$$

$$\frac{\partial L}{\partial b} = \frac{1}{n} \sum (\hat{y} - y)$$

Iteration 1

$$\frac{\partial L}{\partial w} = 0.1669$$

$$\frac{\partial L}{\partial b} = 0.5667$$

update, weight

$$w = 3 - 0.1 \times 0.1669 = 2.98$$

$$b = 1 - 0.1 \times 0.566 = 0.94$$

Iteration 2

$$\frac{\partial L}{\partial w} = 0.16$$

$$\frac{\partial L}{\partial b} = 0.56$$

update,

$$w = 2.98 - 0.1 \times 0.16 = 2.96$$

$$b = 0.94 - 0.1 \times 0.56 = 0.88$$

Answer the Question no-8

a) $w_1 = 0.2$

$w_2 = 0.3$

$b = 0.1$

$\alpha = 0.05$

$\hat{y} = \sigma(w_1 x_1 + w_2 x_2 + b)$

<u>Sample</u>	<u>Actual y</u>	<u>predicted $\hat{y}$</u>
1	0	0.90
2	1	0.79
3	0	0.90
4	1	0.87

$BCE = -\frac{1}{n} \sum [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$

$= 1.289$

$$b) \frac{\partial L}{\partial w_1} = \frac{1}{n} \sum (\hat{y} - y) x_1$$

$$\frac{\partial L}{\partial w_2} = \frac{1}{n} \sum (\hat{y} - y) x_2$$

$$\frac{\partial L}{\partial b} = \frac{1}{n} \sum (\hat{y} - y)$$

Iteration 1

$$\frac{\partial L}{\partial w_1} = 2.05$$

$$\frac{\partial L}{\partial w_2} = 1.55$$

$$\frac{\partial L}{\partial b} = 0.37$$

updated values,

$$w_1 = 0.2 - 0.05 \times 2.05 = 0.09$$

$$w_2 = 0.3 - 0.05 \times 1.55 = 0.22$$

$$b = 0.1 - 0.05 \times 0.37 = 0.08$$

Iteration 2

$$\frac{\partial L}{\partial w_1} = 1.29$$

$$\frac{\partial L}{\partial b} = 0.25$$

$$\frac{\partial L}{\partial w_2} = 1.34$$

updated values,

$$w_1 = 0.09 - 0.05 \times 1.29 = 0.03$$

$$w_2 = 0.22 - 0.05 \times 1.34 = 0.15$$

$$b = 0.08 - 0.05 \times 0.25 = 0.068$$

c) $x_1 = 4$, $x_2 = 1$

$$z = 0.03 \times 4 + 0.15 \times 1 + 0.068$$
$$= 0.3546$$

$$\hat{y} = \sigma(0.3546) = 0.5878$$

∴ $0.5878 > 0.5$

∴ Final prediction,

Has Disease = Yes

∴ probability = 58.78%

Answer the Question no-9

Given, $x_1 = 5$

$$x_2 = 1$$

Hidden neuron h_1 ,

$$x_1 \rightarrow h_1 = 4$$

$$x_2 \rightarrow h_1 = -1$$

$$z_1 = 5 \times 4 + 1 \times (-1) = 19$$

$$h_1 = \max(0, 19) = 19 \text{ (ReLU)}$$

Hidden neuron h_2 ,

$$x_1 \rightarrow h_2 = 3$$

$$x_2 \rightarrow h_2 = 2$$

$$z_2 = 5 \times 3 + 1 \times 2 = 17$$

$$h_2 = \max(0, 17) = 17 \text{ (ReLU)}$$

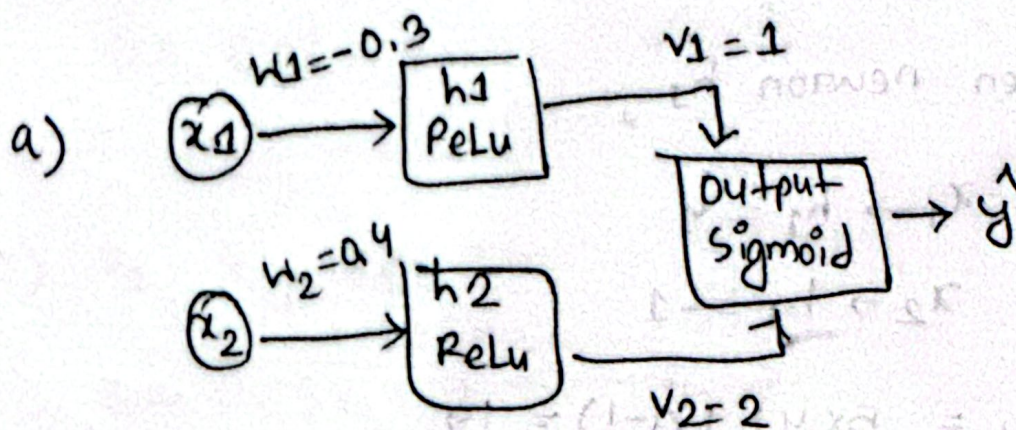
Output layer,

$$z = 19 \times 3 + 17 \times 2 = 91$$

$$\sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-91}} \approx 1$$

Answer the Question no-10

Answer the Question no-10



b) Forward Propagation,

Hidden neuron 1,

$$z_1 = x_1 w_1 = 1 \times -0.3 = -0.3$$

$$h_1 = \text{ReLU}(-0.3) = 0$$

$$z_2 = 1 \times 0.4 = 0.4$$

$$h_2 = \text{ReLU}(0.4) = 0.4$$

$$z_0 = 0 \times 1 + 0.4 \times 2 = 0.8$$

$$\hat{y} = \frac{1}{1 + e^{-0.8}} \approx 0.69$$

c) $y = 1$

$$L = -[y \log y^1 + (1-y) \log (1-y^1)]$$

$$= -\log(y^1) = -\log(0.689)$$

$$= 0.371$$

d) Backpropagation and weight Updates ,

$$\alpha = 0.1$$

$$\frac{\partial L}{\partial z_0} = y^1 - y = 0.68 - 1 = -0.31$$

For v_1 ,

$$\frac{\partial L}{\partial v_1} = -0.31 \times 0 = 0$$

$$v_1(\text{new}) = v_1 - \alpha \frac{\partial L}{\partial v_1} = 1 - 0.1 \times 0 = 1$$

For v_2 ,

$$\frac{\partial L}{\partial v_2} = \frac{\partial L}{\partial z_0} \cdot h_2 = -0.31 \times 0.4 = -0.12$$

$$v_2(\text{new}) = 2 - 0.1 \times (-0.12) = 2.01$$

Hidden layer weight update,

$$L = 0.1$$

For w_1 ,

$$z_1 = -0.3 \quad \text{ReLU}'(z_1) = 0$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z_0} \cdot v_1 \cdot \text{ReLU}'(z_1) \cdot x_1$$

$$= -0.31 \times 1 \times 0 \times 1 = 0$$

$$w_1^{\text{new}} = -0.3 - 0.1 \times 0 = -0.3$$

For w_2 ,

$$z_2 = 0.4$$

$$\text{ReLU}'(z_2) = 1$$

$$\frac{\partial L}{\partial w_2} = -0.31 \times 2 \times 1 \times 1 = -0.62$$

$$w_2^{\text{new}} = 0.4 - 0.1 \times (-0.62) = 0.46$$

Answer the Question no-11

a) For binary classification,

$$L = -[y \log(y') + (1-y) \log(1-y')]$$

true label, $y=1$,

$$\therefore L = -\log(y')$$

b) Given,

$$x_1 = 1, x_2 = 2$$

$$w_1 = 0.1, w_2 = 0.5, w_3 = 0.5, w_4 = 0.2$$

$$\beta_1 = \beta_2 = \beta_3 = 0.5$$

perception, P_1

$$z_1 = 0.1 \times 1 + 0.5 \times 2 + 0.5 = 1.6$$

$$a_1 = \sigma(1.6) = 0.83$$

perception, P_2

$$z_2 = 0.5 \times 0.83 + 0.5 = 0.91$$

$$a_2 = \sigma(0.91) = 0.71$$

perception p_3

$$z_3 = w_4 a_2 + \beta_3$$

$$= 0.2 \times 0.71 + 0.5 = 0.64$$

$$y' = a_3 = \sigma(0.64) = 0.65$$

BCE error,

$$L = -\log(0.65) = 0.4225$$

c) Backward propagation

$$\frac{\partial L}{\partial z_3} = y' - y = 0.65 - 1 = -0.34$$

$$z_3 = w_4 a_2 + \beta_3$$

$$\frac{\partial L}{\partial w_4} = (y' - y) a_2$$

$$= -0.34 \times 0.71$$

$$= -0.24$$

$$\frac{\partial L}{\partial w_1} = (y' - y) w_4 a_2 (1 - a_2) w_3 a_1 (1 - a_1) x_1$$

$$= -0.34 \times 0.2 \times 0.71 (1 - 0.7) 0.5 \times 0.83$$

$$= -0.00098$$